

Break Down the Problem

In the space below, define the problem statement:

Develop a MATLAB program that models and analyzes a lithium battery charging profile using the voltage, power and current vs time data provided to us. This project aims to model a battery charging as a simple first-order RC system, then asks us to estimate a tau value ($R \cdot C$) by generating a fitted charging-curve model, create three subplots with their respective y-axis values, and perform graphical analysis to meet the designated guidelines.

List the requirements, constraints, and success criteria for the project:

A parameter that constrains our project model is the energy that is lost to heat in the battery (due to resistance). Looking at this question another way, an implied constraint/requirement to our project is that we only need to consider the charging portion of the battery charging data cycles, not the discharging portion. The size of the data table provided is also a constraint in a sense because there is a substantial amount of data, whereas for some task steps we only need to consider one data cycle.

What is your proposed solution?

Create a MATLAB live script that imports and analyzes the given battery charging data, generates the appropriate graphs for visualization and performs numerical analysis using integration and differentiation to answer all inquiries needed to fulfill the project's criteria.

Additionally, we plan to incorporate beneficial features to improve our submission, including but not limited to: adding the option to choose which cycle is to be analyzed, splitting the data into

two regions (CC and CV phases), and using Simscape Battery to model/compare various charging profiles.

List the steps you will need to take to achieve your proposed solution:

1. Import and organize the lithium-ion battery dataset.
2. Utilize MATLAB's curve fitting toolbox to compare different charging cycles at various times using multiple models.
3. Plot the battery's voltage data alongside with any additional data such as indexes, capacities or residuals.
4. Generate voltage, current and power vs time subplots with appropriate axes, titles and legends.
5. Compute the derivative of the voltage data using the gradient function.
6. Calculate the time required for the battery to reach 80% and 100% charge.
7. Compute the total energy delivered to the battery using integration to calculate the area under the power vs time curve.
8. Estimate the energy loss through the battery's internal resistance that's converted to heat.
9. Summarize findings and analytical results into a MATLAB table.

What challenges do you face in the process of achieving the solution?

Some challenges that will be faced include researching the properties and behavior of lithium-ion batteries over time, processing a large dataset and finding a way to only extract one desired cycle

from it, and understanding the equations and properties of an RC circuit used to model the battery charging profile.